第 01 章 — 一次工具调用

TL;DR

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为什么这很重要

“4,892,769 ” “ ”
bug——

——

核心概念

模型负责写请求，你的代码负责干活

——“
”——

“ ”
——

四步循环

```
sequenceDiagram
    participant App as
    participant LLM as
    participant Tool as

    App->>LLM:         +      schema
    LLM-->>App: tool_use
    App->>Tool: execute(name, arguments)
    Tool-->>App:
    App->>LLM: tool_result
    LLM-->>App:
```

1. —— JSON Schema
2. ——Anthropic API `tool_use` OpenAI API `tool_calls` —— `id`
3. schema
4. `id` `tool_result`

```
//    2                --
{ id: "call_abc", name: "get_weather", input: { city: "Tokyo" } }

//    4               --          id  content
{ tool_use_id: "call_abc", content: "18°C, partly cloudy" }
```

shell

Schema 就是契约

- **Name** ——
- **Description** ——
  “ ”
  “ ”
- **schema** **Input schema** —— JSON Schema

```
//            --                        SDK
{
  name: "get_weather",
  description: "                                        \
                                  ",
  input_schema: {
    type: "object",
    properties: { city: { type: "string", description: "     'Tokyo'" } },
    required: ["city"]
  }
}
```

bug

“ X”

Schema 和函数必须同步变化

schema `city` `location`

schema `city` `{ "city": "Tokyo" }`

`location` —— schema

Schema schema

Sebastian Raschka

—— schema

错误输入和故障应该成为消息，而不是异常

schema

schema

——

- schema
- `tool_result`
- `tool_result` ——

工具结果也有自己的结构

`id` `tool_use` `id` `tool_result` `id`

50 KB grep 2 MB

OpenCode Hermes Agent

值得了解的服务商特定控制项

- `tool_choice` “X” `none` Anthropic OpenAI Bedrock Gemini
- `tool_use` OpenAI `parallel_tool_calls: false` 02
- **schema** OpenAI `strict: true` JSON Schema schema JSON Schema
- “ ” “ schema JSON” JSON Schema —— `response_format`
- —— schema
- Anthropic `refusal` OpenAI —— 18

服务商各不相同，概念始终一致

Anthropic `tool_use` `input` OpenAI `tool_calls` `arguments` Bedrock
SDK Vercel AI SDK LangChain Hermes Agent OpenCode
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OpenCode Hermes Agent

真实系统笔记

- **OpenCode** `Tool.define` schema
“
”
- **Hermes Agent** `ToolEntry` schema
- **OpenClaw** **Paperclip** “ ” shell
—— “ +
schema + ”

常见失败情况

——
AI

- `tool_use` ——
“ *tool_use*
” —— *id* `tool_result`
- JSON schema id
schema ——
`tool_result`

03

-
  `tool_choice`
-
  `tool_result` 02
-
  “ ”——

与你的智能体结对

- “ JSON ”
- “ schema ”
- “ `tool_result` ”
- “ 4 KB ”
- “ OpenCode ——”

下一步

02